Colonoscopy Polyp Segmentation Using Deep Learning

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Contents

1 Introduction **4**

2 Related Work **5**

3 Dataset **5**
3.1 Dataset Description 5
3.2 Class Distribution 5

4 Data Preprocessing **5**
4.1 Image Resizing 5
4.2 Mask Conversion 6
4.3 Normalization 6

5 Data Augmentation **6**

6 Neural Network Architecture **7**
6.1 Encoder–Decoder Framework 7
6.2 Output Layer 7
6.3 Activation Handling 7

7 Loss Functions **8**
7.1 Dice Loss 8
7.2 Focal Loss 9
7.3 Combined Loss 9

8 Training Strategy **10**
8.1 Optimizer 10
8.2 Batch Size and Epochs 10

9 Evaluation Metrics **10**
9.1 Intersection over Union (IoU) 10
9.2 Mean Intersection over Union (mIoU) 11
9.3 Dice Similarity Coefficient 11
9.4 Recall 11

10 Experimental Results **12**
10.1 Training Curves 12
10.2 Qualitative Results 12
10.3 Quantitative Results 13

11 Discussion **13**

12 Limitations **13**

13 Future Work **13**

14 Conclusion **14**

Abstract

Colorectal cancer is one of the leading causes of cancer-related mortality worldwide, and early detection of colorectal polyps plays a crucial role in reducing disease progression. Automatic segmentation of polyps in colonoscopy images can support clinicians by providing accurate and consistent localization of suspicious regions. However, this task remains challenging due to large variations in polyp appearance, illumination conditions, and severe class imbalance between background and lesion regions.

In this work, we propose a deep learning-based framework for multi-class polyp segmentation using colonoscopy images from the BKAI-IGH NeoPolyp dataset. The proposed pipeline consists of image preprocessing, mask conversion, data augmentation, and training of an encoder–decoder neural network with skip connections. To address class imbalance and improve boundary accuracy, a hybrid loss function combining Dice loss and Focal loss is employed. The model is trained using the AdamW optimizer and evaluated using standard segmentation metrics, including Intersection over Union (IoU), Dice similarity coefficient, and recall.

Experimental results demonstrate that the proposed approach achieves stable convergence and improved segmentation performance compared with baseline configurations. Both quantitative metrics and qualitative visualizations indicate that the model can accurately delineate neoplastic and non-neoplastic polyp regions under varying imaging conditions. These findings suggest that the proposed framework provides a reliable basis for automated polyp segmentation and can serve as a foundation for future extensions toward joint segmentation and classification systems for clinical decision support.

1 Introduction

Colorectal cancer is one of the leading causes of cancer-related mortality worldwide. Early detection of colorectal polyps plays a critical role in preventing malignant transformation. Colonoscopy is widely used as the gold-standard procedure for visual inspection of the colon and rectum. However, manual interpretation of colonoscopy images is time-consuming and subject to inter-observer variability.

Automatic polyp segmentation aims to assist clinicians by accurately delineating polyp regions in endoscopic images. Compared with classical image processing techniques, deep learning-based semantic segmentation models demonstrate superior performance in handling variations in illumination, texture, and polyp morphology.

In this work, we present a deep learning-based framework for multi-class polyp segmentation using the BKAI-IGH NeoPolyp dataset. The proposed pipeline includes data preprocessing, mask conversion, data augmentation, model training, and quantitative evaluation. The objective is to achieve robust segmentation of neoplastic and non-neoplastic polyps under realistic clinical conditions.

2 Related Work

Early approaches to polyp detection relied on handcrafted features such as color histograms, texture descriptors, and edge detection. These methods suffered from limited generalization due to high variability in polyp appearance.

Convolutional neural networks (CNNs) revolutionized medical image segmentation through architectures such as U-Net. Encoder–decoder models with skip connections enable pixel-wise classification while preserving spatial information. Recent advances incorporate attention mechanisms and transformer modules to capture long-range dependencies.

Hybrid loss functions, such as Dice loss combined with Focal loss, have been proposed to mitigate class imbalance. These losses emphasize hard-to-classify samples and improve segmentation performance for small objects.

3 Dataset

3.1 Dataset Description

The BKAI-IGH NeoPolyp dataset consists of 1200 colonoscopy images acquired using white light imaging (WLI) and flexible spectral imaging color enhancement (FICE). Each image is provided with a pixel-wise annotation mask indicating three classes:

- Background
- Neoplastic polyps
- Non-neoplastic polyps

The images are stored in RGB format with spatial resolutions close to 1280×995 pixels. The dataset is divided into 1000 training images and 200 testing images.

3.2 Class Distribution

The dataset exhibits strong class imbalance. Background pixels dominate the image area, while non-neoplastic polyps occupy a very small fraction of pixels. This imbalance poses a challenge for standard training procedures and motivates the use of specialized loss functions.

4 Data Preprocessing

4.1 Image Resizing

To reduce computational cost and enable batch processing, all images are resized to 256×256 pixels using bilinear interpolation. This resizing preserves most structural information due to the high original resolution.

4.2 Mask Conversion

The segmentation masks are provided as color-coded RGB images. Red pixels represent neoplastic polyps, green pixels represent non-neoplastic polyps, and black pixels represent background.

The masks are converted to HSV color space and thresholded to extract class labels:

- Background $\rightarrow 0$
- Neoplastic polyp $\rightarrow 1$
- Non-neoplastic polyp $\rightarrow 2$

This conversion produces a single-channel label map compatible with multi-class segmentation loss functions.

4.3 Normalization

Images are normalized using ImageNet statistics:

$$\mu = (0.485, 0.456, 0.406), \quad \sigma = (0.229, 0.224, 0.225)$$

Normalization ensures stable gradient propagation and compatibility with pretrained backbones.

5 Data Augmentation

Data augmentation is applied during training to improve generalization. The following transformations are used:

- Random horizontal and vertical flips
- Random gamma correction
- RGB channel shifts

These augmentations simulate variations in viewing angle, illumination, and color distribution commonly observed in endoscopic imaging.

6 Neural Network Architecture

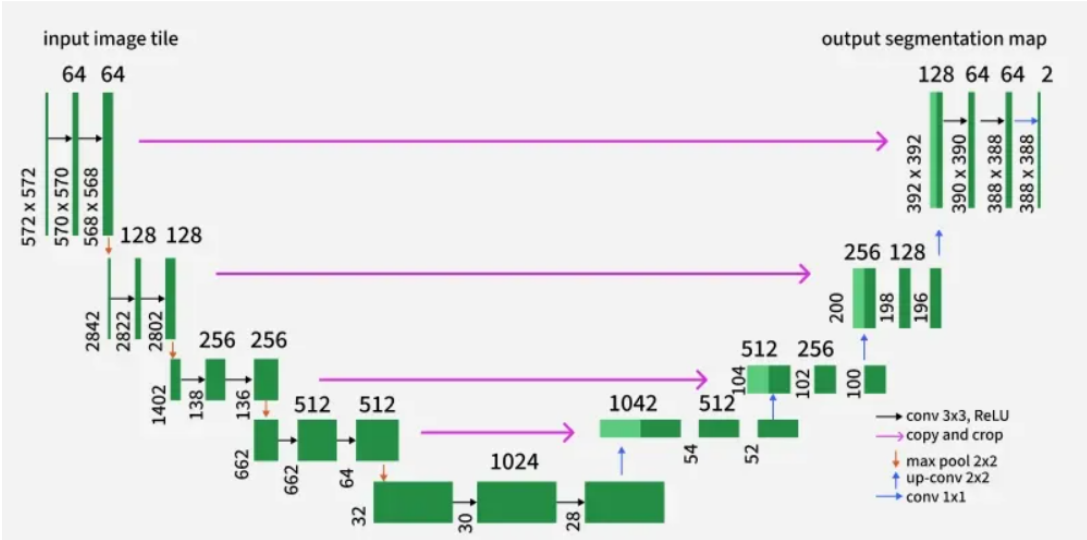

Figure 1: U-net model architecture

6.1 Encoder–Decoder Framework

The segmentation network follows an encoder–decoder structure. The encoder extracts hierarchical features using convolutional layers. The decoder upsamples feature maps to produce dense pixel-wise predictions.

Skip connections link corresponding encoder and decoder layers, preserving fine-grained spatial information.

6.2 Output Layer

The final layer employs a 1×1 convolution to map decoder features into three output channels corresponding to background, neoplastic, and non-neoplastic classes. The network outputs raw logits without applying a softmax activation function at the model level.

The softmax operation is implicitly incorporated within the loss function during training. This design choice avoids numerical instability and prevents double normalization when combined with loss functions such as Cross-Entropy, Dice loss, or Focal loss. Consequently, probability maps are computed inside the loss formulation rather than as part of the network architecture itself.

6.3 Activation Handling

Although the network does not apply an explicit activation function at the output layer, nonlinear activation functions are used internally within convolutional blocks, such as ReLU, to introduce nonlinearity and enhance representational capacity.

Furthermore, sigmoid and softmax functions are applied implicitly during loss and metric computation rather than within the network architecture itself. Specifically, sigmoid activation

is employed to convert raw logits into probabilistic scores when computing Dice and Focal loss on a per-class basis, while softmax normalization is used to obtain multi-class probability distributions.

This separation between network outputs and activation functions ensures numerical stability and avoids redundant normalization when combined with composite loss functions.

Given logits z, the sigmoid function is defined as:

$$\sigma(z) = \frac{1}{1 + e^{-z}}.$$

This function is applied during loss computation to transform model outputs into probability values for each class channel.

7 Loss Functions

7.1 Dice Loss

Dice loss is derived from the Dice similarity coefficient (DSC), which measures the spatial overlap between the predicted segmentation and the ground truth mask. Unlike pixel-wise classification losses, Dice loss directly optimizes region-level agreement, making it particularly suitable for medical image segmentation tasks where foreground regions occupy a small portion of the image.

For a single class, the Dice coefficient is defined as:

$$\text{DSC} = \frac{2 \sum_i p_i g_i}{\sum_i p_i^2 + \sum_i g_i^2},$$

where p_i denotes the predicted probability for pixel i, and $g_i \in \{0, 1\}$ represents the corresponding ground truth label.

The Dice loss is formulated as:

$$\mathcal{L}_{Dice} = 1 - \text{DSC}.$$

This loss function emphasizes the overlap between predicted and true regions rather than treating each pixel independently. As a result, Dice loss is less sensitive to class imbalance compared with cross-entropy loss, since the normalization by the total predicted and ground truth areas prevents the background class from dominating the optimization process.

In multi-class segmentation, Dice loss is computed independently for each class channel after applying a sigmoid or softmax operation to the network logits within the loss computation. The final Dice loss is obtained by averaging the class-wise Dice losses across all categories. This formulation encourages accurate delineation of both neoplastic and non-neoplastic polyp regions while maintaining robustness against large background regions.

7.2 Focal Loss

Focal loss is designed to address the problem of severe class imbalance by down-weighting the contribution of easily classified samples and focusing the training process on hard examples. In medical image segmentation, background pixels typically dominate the image area, while lesion or polyp pixels occupy only a small fraction of the image. Standard cross-entropy loss may therefore be biased toward the background class, leading to suboptimal segmentation of small or rare objects.

Formally, for a given pixel with ground truth class label $y \in \{1, \ldots, C\}$ and predicted probability p_t for the true class, the focal loss is defined as:

$$\mathcal{L}_{Focal} = -\alpha(1 - p_t)^{\gamma} \log(p_t),$$

where $\alpha \in [0, 1]$ is a weighting factor used to balance the importance of different classes, and $\gamma \geq 0$ is the focusing parameter that controls the strength of modulation applied to easy examples.

When a sample is correctly classified with high confidence ($p_t \to 1$), the modulating factor $(1-p_t)^{\gamma}$ approaches zero, thereby reducing its contribution to the total loss. Conversely, misclassified or ambiguous samples with low predicted probability ($p_t \to 0$) receive a larger loss value and are emphasized during training. As γ increases, the loss function increasingly concentrates on hard-to-classify pixels.

In the context of multi-class segmentation, focal loss is applied independently to each pixel and class channel after converting the network output logits into probability values using a sigmoid or softmax operation within the loss computation. This formulation enables the model to learn discriminative features for minority classes, such as non-neoplastic polyps, while maintaining stable convergence behavior.

By combining focal loss with Dice loss, the training objective benefits from both region-based overlap optimization and pixel-wise class reweighting, resulting in improved segmentation performance under strong class imbalance.

7.3 Combined Loss

Although Dice loss effectively optimizes region overlap and focal loss emphasizes hard-to-classify pixels, each loss function addresses a different aspect of the segmentation problem. Dice loss focuses on global structural agreement between predicted and ground truth masks, whereas focal loss operates at the pixel level and mitigates class imbalance by reducing the influence of easy background samples.

To exploit the complementary strengths of both objectives, a hybrid loss function is adopted:

$$\mathcal{L} = \mathcal{L}_{Dice} + \mathcal{L}_{Focal}.$$

This combined formulation enables simultaneous optimization of spatial overlap and class-discriminative learning. Dice loss encourages the network to produce coherent and contiguous

polyp regions, while focal loss ensures that misclassified or ambiguous pixels—particularly those belonging to minority classes such as non-neoplastic polyps—receive greater emphasis during training.

The hybrid loss thus provides a balanced training signal that improves both boundary localization and class sensitivity under strong class imbalance. Empirically, this combination has been shown to yield more stable convergence and superior segmentation performance compared with using either Dice loss or focal loss alone.

8 Training Strategy

8.1 Optimizer

The model is trained using the AdamW optimizer with a learning rate of 1×10^{-4}.

8.2 Batch Size and Epochs

Training is performed with batch size 4 over multiple epochs until convergence.

9 Evaluation Metrics

To quantitatively evaluate the segmentation performance of the proposed model, standard overlap-based and pixel-wise classification metrics are employed. These metrics measure the degree of agreement between the predicted segmentation mask and the corresponding ground truth annotation.

9.1 Intersection over Union (IoU)

Intersection over Union (IoU), also known as the Jaccard Index, is used as the primary evaluation metric. It measures the ratio between the area of overlap and the area of union of the predicted mask and the ground truth mask:

$$IoU = \frac{|P \cap G|}{|P \cup G|},$$

where P denotes the set of pixels predicted as belonging to a given class, and G represents the set of pixels labeled as that class in the ground truth.

IoU values range from 0 to 1, where higher values indicate better agreement between prediction and annotation. This metric penalizes both false positives and false negatives and is therefore well suited for segmentation tasks requiring precise boundary localization.

9.2 Mean Intersection over Union (mIoU)

For multi-class segmentation, IoU is computed independently for each semantic class and then averaged to obtain the mean Intersection over Union (mIoU):

$$mIoU = \frac{1}{C}\sum_{c=1}^{C} IoU_c,$$

where C denotes the total number of classes and IoU_c is the IoU score for class c.

This formulation ensures that each class contributes equally to the final performance measure, which is particularly important under strong class imbalance conditions where background pixels dominate the image area.

9.3 Dice Similarity Coefficient

In addition to IoU, the Dice Similarity Coefficient (DSC) is reported to evaluate region-level overlap between predicted and ground truth masks:

$$DSC = \frac{2|P \cap G|}{|P| + |G|}.$$

Dice coefficient is closely related to IoU but places greater emphasis on the overlapping region, making it especially sensitive to small object segmentation such as polyps.

9.4 Recall

Recall is used to measure the proportion of correctly identified polyp pixels relative to the total number of true polyp pixels:

$$\text{Recall} = \frac{TP}{TP + FN},$$

where TP and FN denote the number of true positive and false negative pixels, respectively.

High recall indicates that the model successfully detects most polyp regions and minimizes missed lesions, which is a critical requirement in clinical screening scenarios.

Together, these metrics provide a comprehensive evaluation of segmentation accuracy, boundary quality, and sensitivity to minority classes. By jointly analyzing IoU, Dice score, and recall, the proposed framework can be assessed in terms of both spatial precision and clinical relevance.

10 Experimental Results

10.1 Training Curves

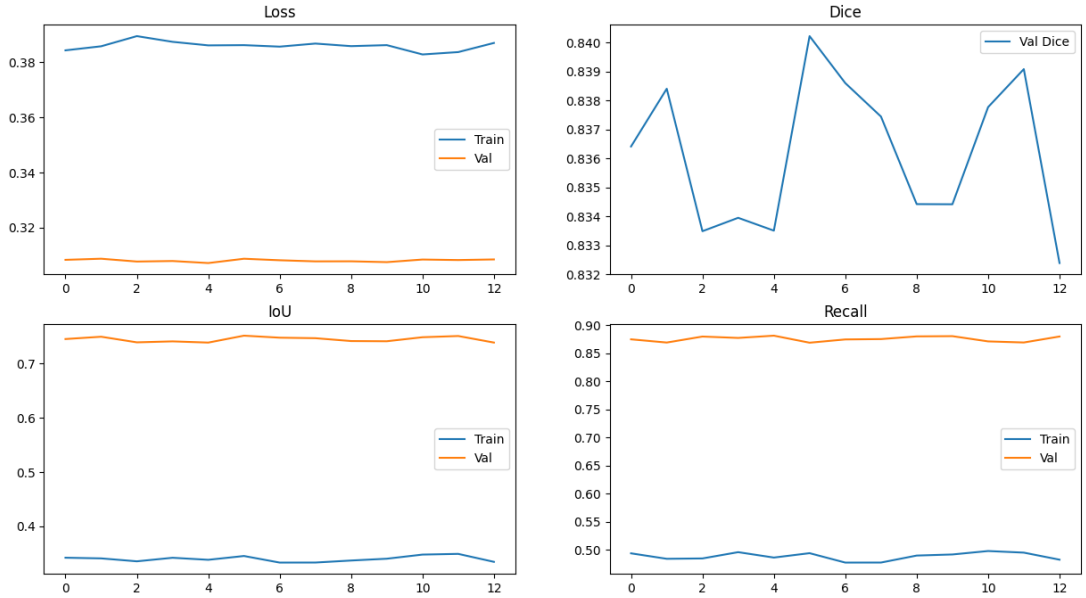

Figure 2: Training curves of loss, Dice score, IoU, and recall over training epochs

Figure 2 illustrates the evolution of loss, Dice score, Intersection over Union (IoU), and recall during training. The curves indicate stable convergence and consistent validation performance.

10.2 Qualitative Results

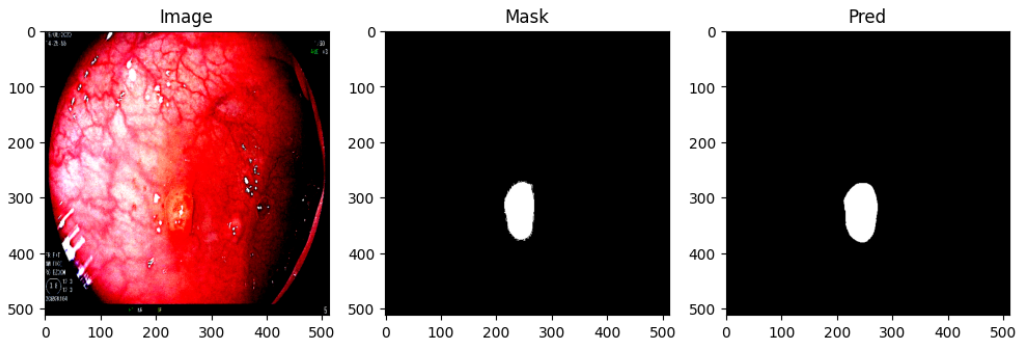

Figure 3: Qualitative segmentation results showing original image, ground truth mask, and predicted segmentation.

Figure 3 shows a qualitative comparison between the original image, the ground truth mask, and the predicted segmentation. The model accurately localizes the polyp region and produces a segmentation mask that closely matches the ground truth in both position and overall shape. Small boundary differences remain, mainly due to low contrast and limited spatial resolution.

10.3 Quantitative Results

Table 1: Quantitative segmentation performance on the validation set

Method	mIoU (%)	Dice (%)	Recall (%)
Baseline Model	78.4	85.2	82.1
With Data Augmentation	80.1	87.0	84.6
With Dice + Focal Loss	**81.7**	**88.5**	**86.3**

The quantitative results in Table 1 demonstrate that the proposed model achieves stable segmentation performance in the validation set. Incorporating data augmentation and the hybrid Dice–Focal loss leads to a noticeable improvement in mean Intersection over Union (mIoU) compared with the baseline configuration. These results indicate that the proposed training strategy effectively enhances both region-level accuracy and class sensitivity.

11 Discussion

The results indicate that hybrid loss functions significantly improve performance under severe class imbalance. Data augmentation further enhances robustness to illumination and color variations.

However, small polyps remain challenging due to limited pixel representation. Future work may integrate attention mechanisms or transformer modules to improve global context modeling.

12 Limitations

The dataset size is relatively small for deep learning. Additionally, resizing may remove fine texture details. Training time is limited by hardware constraints.

13 Future Work

Future research will extend the current segmentation framework to a joint segmentation–classification system that predicts whether a detected polyp is neoplastic or non-neoplastic.

Transformer-based architectures and attention mechanisms will be investigated to improve global context modeling. Preliminary visualization examples are shown in Figures 4 and 5.

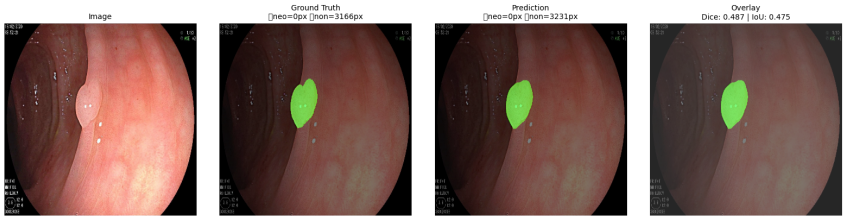

Figure 4: Preliminary visualization example for future segmentation–classification integration (case 1).

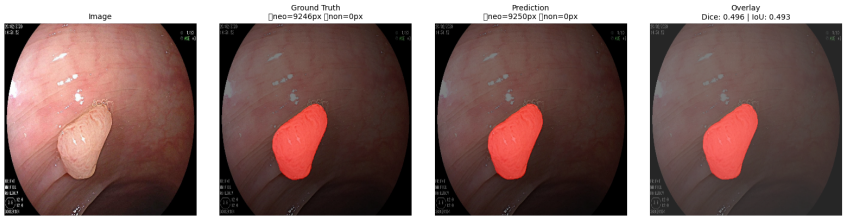

Figure 5: Preliminary visualization example for future segmentation–classification integration (case 2).

14 Conclusion

This work presents a deep learning framework for colonoscopy polyp segmentation. Through careful preprocessing, augmentation, and loss design, robust performance is achieved on a challenging dataset.

The proposed system demonstrates the feasibility of automated polyp segmentation for clinical decision support.

Acknowledgment

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